

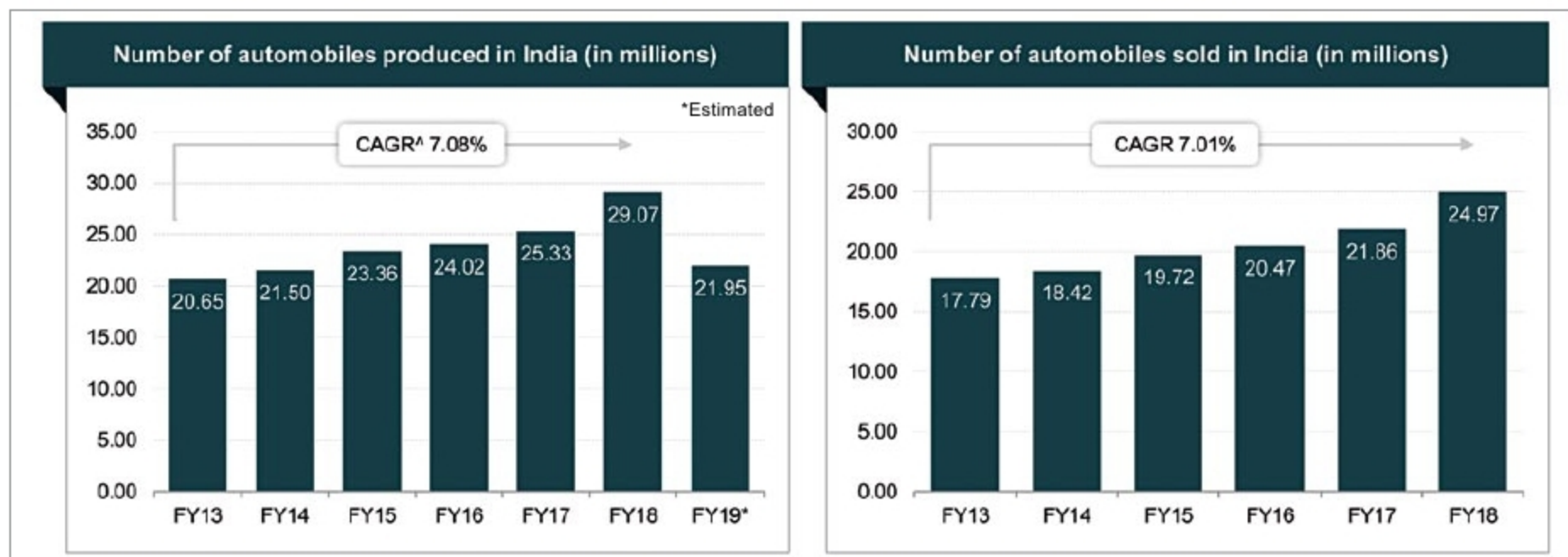


Fig. 1: Indian automotive market overview [Source: Society of Indian Automobile Manufacturers (SIAM), IBEF]

turing) is expected to reach ₹ 16.16 to 18.18 trillion (US\$ 251.4 to 282.8 billion) by 2026. The Indian automotive industry is also targeting to increase exports of vehicles five times during the period 2016 to 2026.

Unprecedented growth of the Indian automotive sector is mainly supported by four factors: favourable policy, growing demand, rising investment and marketing opportunities. The Automotive Mission Plan: 2016-26 indicates a clear vision of the government for this sector. The government aims to develop India as a global automotive manufacturing centre.

Moreover, India has significant cost advantages that result in savings of around 10 to 25 per cent on operations by Indian automotive manufacturers vis-à-vis those in Europe and Latin-America.

The cumulative FDI inflow in the automotive sector was around US\$ 19.29 billion during the period April 2000 to June 2018. The Central government expects this sector to attract US\$ 8 to 10 billion in local and foreign investments by 2023.

Market mix

The Indian automotive market can be bucketed under the following broad categories (Fig. 2):

- Two-wheelers
- Passenger vehicles
- Commercial vehicles
- Three-wheelers

Two-wheelers and passenger vehicles dominate the domestic Indian auto market. Passenger car sales are

further governed by small- and mid-size cars. Two-wheelers and passenger cars accounted for 81 and 13 per cent of over 24.97 million vehicles sold in FY18, respectively. Overall automotive exports reached 4.04 million vehicles in FY18. Two-wheelers made up 69.7 per cent of the total exported vehicles, followed by passenger vehicles at 18.5 per cent, three-wheelers at 9.4 per cent and commercial vehicles at 2.4 per cent.

Green mobility, the game changer

Continuous regulatory pressures to stop environmental hazards associated with internal combustion engines (ICEs) coupled with availability of advanced technologies for electric power trains and storage systems (batteries, etc) ensure better functionalities and enhanced demand for electric vehicles (EVs) worldwide.

According to Reportbuyer, the global EV market is growing at a CAGR of 21.27 per cent for the forecast period of 2017-2026. The market is driven by growing emissions of carbon-dioxide and greenhouse gases, availability of low-cost lithium-ion batteries and various government initiatives that are encouraging the use of EVs.

The Indian automotive sector is also shifting its focus to EVs and other sustainable options. India could be a leader in shared mobility by 2030, providing opportunities for EVs and autonomous vehicles. Innovation is likely to intensify among engine technology and alternative fuels. The government aims to build India into an R&D hub,

and will set up incubation centres for startups working in the EV space.

According to a McKinsey report, global sales of pure battery EVs (excluding hybrids) grew by approximately 45 per cent in 2016. Thus, most of the car manufacturers have started going beyond ICE legacies while introducing new models.

Rise of electronics

The world is seeing electronics permeate all areas. The automotive industry is no different. Current penetration of electronics in the automotive market is low in India, due to limited scale—imports address around 65 to 70 per cent of OEM demand in the country. But by 2030, auto electronics content is expected to contribute nearly 45 per cent of the total automobile cost in India.

Rapidly evolving emissions and safety regulations as well as technological disruptions such as connectivity and e-mobility could underpin the demand for electronics at OEM and customer levels. For instance, it is expected that implementation of BS-VI standards will lead to a spike in demand for components like catalytic converters, electronic fuel injection systems, oxygen sensors and intelligent battery sensors.

EVs need a higher number of motor controllers than their counterparts. Each EV consists of at least a charger and a battery management system on board, both using power electronics. Moreover, all EVs need electric suspension with more involvement of power electronic devices and components, including power management semi-